

# GestaltX

An iterative, gap-driven research assistant for the Ashen Era Archive

SUBMISSION REPORT · 9 SEPTEMBER 2026

Demonstration video (unlisted): [insert YouTube URL here]

## 1 · Problem Statement

Enterprise documentation is messy: scanned pages, diagrams, tables and plain text spread across hundreds of documents. The information a user needs is present, but no single passage contains it. The competition corpus — the Ashen Era Archive — reproduces this faithfully: 415 documents and roughly 1,277 pages across novels, wiki articles, codex data books and in-world ephemera. The same characters, factions, events and artifacts recur throughout, yet no one document tells the whole story, and sources differ in reliability: a tavern ballad and an official codex entry do not always agree.

Sub-track 1C asks for an assistant that searches the way a human expert does: decide where to look, read what is found, recognise what is still missing, and let each finding guide the next search — repeating until the evidence supports a complete answer.

## 2 · Chosen Sub-track — 1C, Searching the Way a Human Does

We chose 1C because it exposes the true weakness of naive retrieval. Questions such as “*When was Gloamreach founded?*” look trivial but are contested inside the archive: a community wiki entry carries a disputed year, the authoritative figure sits in a codex plate reached only by following a documentary pointer, and a near-name decoy (*Gloammarch*) supplies a plausible but wrong answer. Any system that fetches a few relevant passages retrieves only a partial picture. GestaltX treats uncertainty as another research step, not as noise to average over. It plans searches, reads evidence, follows pointers, arbitrates conflicts by source authority, and writes a cited brief a non-technical reader can follow — with the full research path visible live in the interface.

## 3 · Solution Overview

GestaltX is a local-first system: a FastAPI backend performs indexing, iterative research and

synthesis; a Next.js client renders the research trace and the answer. Hybrid retrieval (SQLite FTS5 for exact names and dates, a NetworkX entity graph for archive relationships, optional dense vectors for paraphrases) feeds a bounded research loop in which a gap critic decides whether the evidence is sufficient. An in-process corpus watcher re-indexes added, edited or deleted files before the next question is answered, so the archive can change under a running system.

*[Figure 1 — System architecture: Authority decisions, indexing and synthesis live in the backend; the client is a pure presentation layer.]*

## 4 · Runtime Path

A question arrives at

POST /api/ask

or the streamed

GET /api/ask/stream

. The runtime first refreshes the corpus: if the watcher detected changes, only the affected files are parsed, chunked and re-indexed, the entity graph is rebuilt, and cached searchers are invalidated. The planner then frames the entity and claim under research and proposes opening queries. Each search round is judged by the gap critic; if evidence is missing, contested, or a document points elsewhere (“*consult the Codex*”), the loop reformulates and searches again. When sufficient, claims are ranked by source authority — codex, then wiki, then chronicles, then ephemera — with conflicts preserved rather than hidden. The answer engine assembles a dossier, writes the brief, validates voice, and streams it section by section.

*[Figure 2 — The research loop: Retrieved gaps, pointers and decoys become the next query instead of the final answer.]*

## 5 · The Answer the Finder Sees

The response is a brief, not a model dump. Five sections: **Answer** — the fact, with the winning document named by friendly title and cited; **What the documents say** — each consulted work, its kind, and its claim; **Why this is the answer** — rival years attributed to named documents and the reason the official source stands; **How this answer was found** — the research path in ordinary language; and **Sources** — dossier rows with locator and role (settles, disputes, pointer, mentions). Voice rules are enforced programmatically: no first person, no file paths in prose, no trust-me filler. Any model output that fails the check is discarded and the heuristic brief is used instead.

## 6 · Walk-through: the Gloamreach Question

The wiki marks Gloamreach's founding as contested. GestaltX reads the entry, treats the contested year as a gap rather than an answer, follows the pointer to the Codex, checks the near-name decoy (Gloammarch, 321 AS) and rejects it as an entity mismatch, then answers **246 AS** with the codex plate cited and the rival year attributed to the wiki. Every step appears as a tool card or a short reasoning note in the live trace. Dropping a new file into the archive folder is enough for the next question to consider it.

## 7 · Key Technical Decisions

- **Iterative research, not one-shot RAG.** A retrieved contested value is a research gap, not an answer. The loop must follow pointers and search again. Chosen after a one-shot prototype stalled exactly on contested claims.
- **Hybrid retrieval.** FTS5 preserves exact names and dates; dense retrieval finds paraphrases; the entity graph follows archive relationships; a reranker combines them. Dense vectors are optional because a full CPU embedding pass is slow — FTS plus the graph suffice for the 1C goldens.
- **Authority-aware arbitration.** Codex (tier 1) outranks wiki, chronicles and ephemera. A higher-tier entity-matched claim wins, but contradictory evidence remains visible in the dossier so the brief is honest.
- **Just-in-time context.** Tools return compact hits and load individual sections on demand, keeping local-model prompts within budget.
- **SSE activity feed.** Server-sent events fit a one-way research stream. Named events (reasoning, activity, answer\_chunk, answer) let the UI show tool cards and progressive sections without a second protocol.
- **Impersonal, source-forward voice.** The finder is not a developer. Answers never say “we”, never print file paths, and must attribute rival years to named documents. A validator rejects violations; an LLM may polish only the Answer and Why sections, and only after that check passes.
- **Incremental corpus watch, not a rebuild script.** A manifest of path, size and mtime drives incremental FTS replacement, JSONL merge and a full graph rebuild (near-name warnings need a global pass). The question path blocks on a refresh lock so a file dropped one second earlier is not missed.
- **Classification fallback for root drops.** Folder family remains the authority signal; a file outside the known folders is inferred from filename keywords, then content headings, then defaults to mid-trust chronicle — competitive, but unable to override official sources.

## 8 · Engineering Practice

Meaningful atomic commits with a clear history; modular backend packages with separation of concerns; dependency lockfiles and environment management so the project runs from the README alone; API keys loaded only from environment variables and never committed; exponential backoff on HTTP 429 in every external call; and a regression suite (voice rules, corpus-watch add/modify/delete and tiering, 1C goldens) that guards the final week's work.

## 9 · What Works

End-to-end answers on contested, pointer-based and decoy-laden questions, verified by an offline evaluation harness against golden answers. Key regression:

1c\_000

resolves Gloamreach to **246 AS** with a Codex citation while Gloammarch's 321 AS is rejected as an entity mismatch; the Gauntlet forging resolves to 391 AS. Live archive refresh is demonstrated by dropping a file into the corpus folder and watching the manifest version increment and the next answer absorb it. Unit tests cover voice validation, watcher behaviour and goldens; evaluation runs in both local and HTTP modes.

## 10 · What Failed — and What We Changed

The first evaluation on 5 September failed on 1c\_000: the loop stalled on the contested year and was captured by the Gloammarch decoy. Two root causes: the system treated a contested retrieved value as answerable, and nothing followed documentary pointers. The recovery enabled pointer-following to the Codex and Annals and added near-name warnings; the question now resolves correctly. Earlier, a one-shot RAG prototype was abandoned for the same class of failure. Dense-vector-only retrieval was also tried and rejected: it blurred exact dates and names that FTS5 preserves.

## 11 · Limitations

- OCR is brittle on degraded scans, unusual typefaces, marginalia and multi-column pages.
- Free OpenRouter models have variable availability and rate limits; Ollama is preferred for demos, with retries and response caching as backstop.
- Local small models can mishandle subtle contradictions, so the LLM is optional polish only — the heuristic brief is the source of truth when validation fails.
- Authority tiers encode archive conventions; they cannot prove a source is historically true.

- Reported confidence is an evidence-quality indicator, not a calibrated probability.
- New files are added to FTS and the graph only unless dense indexing is explicitly enabled.
- The watcher’s ~1.5 s debounce can delay a file still being copied; the next poll picks it up.
- Root-dropped files without recognisable names or headings become mid-trust chronicles.
- The official sample-question artifact is not distributed with the workspace, so unavailable eval prompts are skipped rather than fabricated.

## 12 · AI Usage Disclosure

**Tools used.** Cursor coding assistants (Composer, Claude- and Grok-family models) for drafting implementations, planning, tests and documentation; Ollama (local) with optional OpenRouter for runtime answer polish. Full conversation logs are exported as plain-text .txt files under `ai_usage/chatlogs/`, and the complete disclosure is at `ai_usage/ai-usage-disclosure.md`.

**Team decisions made by humans.** The iterative loop over one-shot RAG; the authority-tier model; the local-first stack; just-in-time retrieval; the impersonal, source-forward brief; the in-process corpus watcher; the honest documentation of the 5 September failure; and the 246 AS / 391 AS goldens. All AI output was reviewed and tested by the team before acceptance — nothing was submitted unvalidated.

## 13 · Team Members, Roles and Contributions

| Member   | Role                      | Primary contributions                                             |
|----------|---------------------------|-------------------------------------------------------------------|
| [Name 1] | Team lead · research loop | Loop and gap-critic design; pointer following; evaluation harness |
| [Name 2] | Backend engineer          | Ingestion, classification, FTS5/graph indexes, corpus watcher     |
| [Name 3] | Backend engineer          | Arbitration, dossier and answer engine, voice                     |

| Member   | Role              | Primary contributions                                        |
|----------|-------------------|--------------------------------------------------------------|
|          |                   | validation, tests                                            |
| [Name 4] | Frontend engineer | Next.js client, research trace, SSE streaming, corpus status |

Deliverables accompanying this report: the full Git repository with history, README with setup and run instructions, markdown design documentation and diagrams under `docs/`, AI chat logs under `ai_usage/chatlogs/`, and the demonstration video linked on page one.